

Session 9 — Last night: first frames

Practical Astronomy Intensive

6 questions 20 minutes 3 concepts 4 diagnostic

ISO as gain not sensitivity, the 500 rule as a guide not a law, and trails as Earth turning under a fixed camera. Two of these are shot on the night — the histogram and the intervalometer run are the honest checks.

- 1 Your 20-second frame of the Milky Way is too dark. A friend says "just push the ISO to 12800, that makes the sensor more sensitive." Is he right?

exposure-triangle Apply Multiple choice

- a No — just leave the shutter open for two minutes instead, which collects genuinely more light
REVEALS Longer shutter assumed always to mean more light — past the trailing limit on a fixed tripod the stars only smear
- b Yes — higher ISO makes the sensor physically more sensitive, so it catches more light
REVEALS ISO believed to make the sensor more sensitive — it amplifies after capture
- c** No — ISO amplifies the signal after capture, so it brightens the noise right along with the stars; the sensor collected the same photons either way ✓
- d No — instead open the aperture as wide as it goes, which is always the better move
REVEALS Wider aperture assumed always better — wide open costs star sharpness and smears the corners into coma

Only aperture and shutter change how many photons land. ISO is a gain knob applied afterwards, which is why a brighter high-ISO frame is not a better-exposed frame. And note the two wrong 'fixes' — both are real options with real costs: wide open smears the corners, longer than the trailing limit smears the stars.

- 2 The 500 rule says: max seconds before trailing = $500 / \text{focal length}$. On a 20mm lens that gives 25 seconds. How should you treat that number tonight?

nightscapes Apply Multiple choice

- a As applying equally everywhere in the frame, so if the centre is clean the corners are too
REVEALS Trailing rate assumed uniform across the sky — equatorial stars smear fastest, polar stars barely move
- b** As a rough film-era guide — on a modern high-resolution sensor stars trail at roughly half that, which is why the tighter NPF rule exists; shoot it and zoom in to check ✓
- c As exact — it is a rule, so 25 seconds is the precise threshold
REVEALS 500 rule treated as exact rather than a rough film-era guide
- d As a floor you can exceed by lowering the ISO, which keeps the stars from trailing
REVEALS Trailing believed to depend on exposure level — the limit is set by Earth's rotation, not brightness

The rule was calibrated to what looked sharp on a film print. Pixel-peep a 45 MP sensor and half that time already shows elongation. And trailing is set by Earth's rotation, so no exposure setting negotiates with it — past the limit you need tracking. Note too that a star near the pole barely moves in the same 20 seconds that smears one on the celestial equator.

3 You want two hours of star trails. What do you actually shoot, and what will the frame show?

star-trail-imaging

Apply

Multiple choice

- a Hundreds of short frames stacked, aimed anywhere you like — the arcs come out as concentric rings whatever direction you point
REVEALS Concentric circles expected in any direction — only a frame centred near Polaris gives rings
- b One two-hour exposure in bulb mode — the trails need the shutter open the whole time to draw continuously
REVEALS Star trails believed to need one enormous exposure — a two-hour frame fogs on skyglow
- c Hundreds of short frames stacked — and the arcs record the stars themselves slowly moving across the sky
REVEALS Trails believed to show the stars actually moving — it is Earth turning under the camera
- d Hundreds of 30-second frames back to back, blended later in lighten mode — one two-hour exposure just fogs on skyglow; and the arcs are Earth turning under a fixed camera, which is why they curve around the celestial pole ✓

Three of these four get part of it. The technique is short subs blended in lighten mode (skyglow accumulates in a long exposure, the trails do not get brighter); the physics is Earth's rotation, not stellar motion; and the geometry means only a frame centred near Polaris gives rings — aim east or west and the same stars draw near-straight streaks.

4 Frame and shoot one 20-second nightscape — sky over a real foreground — then bring up its histogram and tell the instructor what it says.

nightsapes

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Watch for: camera on the tripod and genuinely still (no hand on it at release — timer or remote); manual mode, focus set on a bright star at live-view magnification rather than left on autofocus or on the infinity mark; a deliberate foreground, not sky alone; exposure at or under the trailing limit for the focal length in use. On the histogram check, the student should read the peak's position — bunched hard against the left means underexposed and the shadows will fall apart when lifted; touching the right means blown highlights. The peak wants to sit off the left wall. Then have them zoom to 100% on a corner and on a star near the pole, and say whether the stars are round — and why the two corners differ. Push back if they propose fixing an underexposure by extending the shutter past the trailing limit.

The histogram is the only honest feedback on a night shoot: the rear screen looks gorgeous in the dark and lies about exposure every time. This is the frame everyone takes home.

- 5 You are shooting a nightscape at 20mm on a fixed tripod and the frame is a stop underexposed. Aperture is already at f/2.8 (the lens opens to f/1.8), shutter is at the trailing limit, ISO is 1600. Which move costs you least?

exposure-triangle

Reason

Multiple choice

- a Raise the ISO a stop, since a more sensitive sensor gathers the missing photons cleanly
REVEALS ISO believed to make the sensor more sensitive rather than amplify after capture
- b** Raise the ISO a stop — it adds noise but keeps the stars round; the shutter is capped by Earth's rotation and opening to f/1.8 would smear the corner stars into coma ✓
- c Open up to f/1.8 — the lens's widest setting is always its best setting
REVEALS Wider aperture assumed always better — wide open costs sharpness and corner coma
- d Double the shutter to 40 seconds — more time is always more light
REVEALS Longer shutter assumed always to mean more light — past the trailing limit the extra seconds only smear each star

The right move and one wrong reason for it both appear here, on purpose — a student who picks the last option has the right hand on the dial and the wrong physics in mind. Every leg of the triangle is already at a hard limit except gain, so gain is where the stop comes from and noise is the price.

- 6 Set up a star-trail sequence: find Polaris by eye, frame it with a foreground, and start an intervalometer run of 30-second subs. Before you walk away, tell the instructor what shape the trails will take and why.

star-trail-imaging

Reason

Practical task

ANSWER / WHAT TO WATCH FOR

Watch for: Polaris found by the student from the Dipper's pointers, not from a phone app; the camera composed so the pole sits in frame with a deliberate foreground; manual focus set on a star; intervalometer running with the gap between subs as short as the camera allows (long gaps break the arcs into dashes — ask why they chose their interval); exposure and ISO fixed manually so subs match. The verbal check is the real assessment: they should say the arcs will be concentric circles centred on the pole BECAUSE the camera is fixed and Earth is turning under it — not because the stars are moving — and should predict that a frame aimed east instead would give near-straight streaks. Bonus if they note the trails near the pole will be short arcs and those farther out long ones, in the same elapsed time.

This closes the camp's circle: the pole star they found by eye on night one is the point every arc turns around on the last night. If they can say why the rings are centred there, the whole week's diurnal-motion material has landed.